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Auditing Data Leakage and Reporting Gaps in EEG Connectivity Deep Learning for Depression Detection

Kunal Chevuri, L. Ileto, Mohammed Aledhari
Prosper High School, Prosper, TX, USA · University of North Texas, Denton, TX, USA

***Abstract*—Deep learning applied to electroencephalographic (EEG) functional connectivity is a prominent approach to detecting major depressive disorder (MDD), with reported accuracies often above 90%. Recent reviews have examined how these studies split data into training and test sets, but not the connectivity pipeline that produces the model input: connectivity estimation, thresholding, normalization, and channel or band selection performed before partitioning can transfer test information into training, and epoch-level partitioning can place segments from one subject on both sides of the split. We define five such leakage surfaces and a four-level risk rating applied from reported methods alone, and applied it to 18 studies. Two raters agreed on 94% of ratings (Cohen's kappa 0.91). Of the audited studies, 50% partitioned at the epoch rather than the subject level, and 22% left at least one preprocessing step undescribed. No study reported fitting connectivity estimation, thresholds, or normalization on the whole dataset; beyond partitioning, the detectable risk came from selecting channels or bands on all data (17% of studies). Only 2 studies rated leakage-free reported an eligible accuracy, too few to test whether accuracy differs by rating. We propose an eight-item reporting checklist that lets a reader verify, from the paper alone, that a connectivity pipeline is free of leakage.**

***Index Terms*—electroencephalography, functional connectivity, deep learning, major depressive disorder, data leakage, cross-validation, reproducibility, brain networks**

I. Introduction

A. Clinical motivation

Major depressive disorder (MDD) is among the leading causes of disability worldwide [1], yet its diagnosis still rests on clinical interviews and self-report scales, which depend on patient cooperation and clinician expertise and have shown limited test-retest reliability [2], [3]. An objective physiological marker would support diagnosis, and electroencephalography (EEG) is an attractive candidate: it is relatively affordable, non-invasive, portable, and resolves neural activity on the millisecond scale [4].

Depression is increasingly understood to involve abnormal communication within and between large-scale brain networks [5]. Functional connectivity—the statistical dependence between signals recorded at different sites, estimated with measures such as coherence, the phase-locking value, or the phase lag index—offers a way to capture that communication from scalp EEG. Deep networks can learn from connectivity matrices directly, and a substantial literature applies them to distinguishing people with MDD from healthy controls, and many studies report classification accuracies above 90% [6].

B. The verification problem

Accuracies of this magnitude would be clinically meaningful if they held. Whether they hold depends on something a reader usually cannot check: whether any information from the test data reached the model during training. This is data leakage, and in neuroimaging machine learning it has been shown to inflate reported performance substantially [7]–[9].

Recent reviews of EEG-based depression detection have examined how studies validate their models—holdout and cross-validation schemes, including per-subject partitioning and leave-one-subject-out validation, and whether models were tested on independent datasets [2], [10]. That is the right question about the split. But a connectivity model does not consume raw EEG. It consumes a connectivity matrix, built through a chain of steps that each run before the split: recordings are segmented into epochs, connectivity is estimated, matrices are thresholded into graphs, values are normalized, and channels or frequency bands are selected. Each step can carry test information into training if it draws on data beyond the training fold. Most papers do not report the order of these steps, which leaves the leakage question unanswerable from the paper alone. Fig. 1 contrasts a leakage-free ordering with a leaking one and marks the five points (L1–L5) at which test information can enter.

Prior reviews audit how these studies split their data. This work audits what happens to the data before it is split. To our knowledge, based on searches of PubMed, OpenAlex, Semantic Scholar and Google Scholar through September 2026, no prior review has audited connectivity-pipeline construction as a leakage surface in this literature.

We ask four questions. How completely do studies report each leakage surface (RQ1)? Can leakage risk be rated reliably from reported methods by independent raters (RQ2)? Is reported accuracy associated with the rating (RQ3)? And, as a secondary question, does reporting completeness change with publication year (RQ4)?

C. Contributions

1) A leakage taxonomy for EEG connectivity pipelines. We identify five points at which construction of a connectivity matrix can allow test information into training, and describe the conditions under which each occurs.
2) A leakage-risk rating scheme. We define a four-level rating applied from reported methods alone, with explicit decision rules, and report inter-rater agreement.

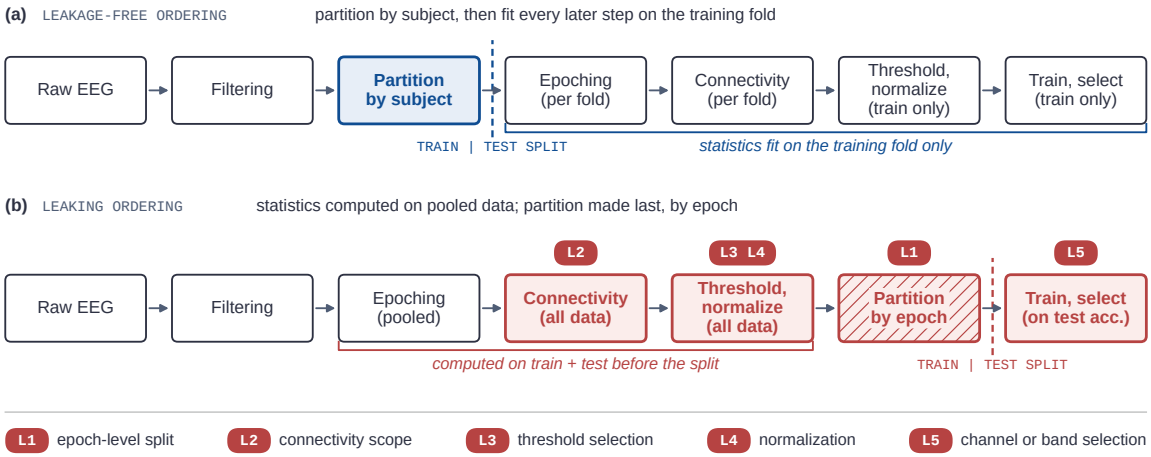


Fig. 1. The connectivity pipeline with leakage surfaces marked. (a) Leakage-free ordering: data are partitioned by subject before any statistic is computed, and every later step, including model and channel or band selection, uses the training fold only. (b) Leaking ordering: connectivity (L2) and threshold or normalization statistics (L3, L4) are computed on pooled data, the partition is made by epoch so one subject can fall on both sides (L1), and channels or bands are chosen by test-set accuracy (L5). The dashed line marks the train/test split; in (b), everything to its left is computed on data from both sides of it. Red boxes and tags mark the leakage points; the hatched box is the epoch-level partition.

3) An audit of the literature. We apply the scheme to 18 studies and report how completely pipelines are described and how reported accuracy varies across risk strata.
4) A reporting checklist. We propose a short set of items that connectivity-based studies should state so leakage can be ruled out by a reader.

II. Related Work

A. Deep learning on EEG connectivity for MDD

In this literature a connectivity measure turns each EEG segment into a channel-by-channel matrix, and a deep network classifies the matrices. Most measures are undirected phase or amplitude couplings (PLV, PLI, coherence, correlation and mutual information) [11]–[15]; a smaller group uses directed measures from multivariate autoregressive models, such as partial directed coherence [16], [17]. Some studies stack one matrix per frequency band into a multilayer network [12], [18].

Convolutional networks treat the matrix as an image, sometimes with a recurrent branch for temporal context [13], [16], [19]; graph neural networks treat channels as nodes and connectivity as edge weights [15], [18], [20], and recent work learns the graph during training [21], [22] or moves it to cortical source space [23]. A few public datasets with tens of participants per group recur (Table I). Reported accuracies commonly exceed 95%, yet these studies seldom state when the pipeline's statistics were computed relative to the train/test split.

B. Existing reviews of validation practice

Recent reviews compare validation schemes in EEG depression studies; one found none that validated across datasets [2], and another found near-perfect accuracy most often where samples were small and validation subject-dependent [10]. One systematic review of EEG-based methods compared preprocessing pipelines, feature extraction and model families, and found that the lack of standardized evaluation protocols limits comparison across studies [6]. A meta-analysis of 176 machine-learning studies of MDD, covering 60,926 participants and several biomarker types, examined the effect of algorithm, validation procedure and comparison group on accuracy; it found that sample size was negatively correlated with reported accuracy and detected publication bias [24]. That association is the confound this audit addresses through sample-size stratification.

The distinction we draw is one of scope. These reviews ask how a study separated training from test data, and whether its validation was subject-dependent or subject-independent. None, to our knowledge, asks whether the connectivity pipeline itself—estimation, thresholding, normalization and selection—was fit before that separation **[PENDING: confirm after reading each review]**. A study can use leave-one-subject-out validation and still leak through a threshold chosen on the pooled data; a review that records only the validation scheme would score it as sound.

C. Data leakage in neuroimaging machine learning

Leakage is well documented in neuroimaging. Across four resting-state fMRI connectome datasets, Rosenblatt et al. found that selecting features on the full dataset and repeating subjects across the split inflated performance most and that effects were more variable in smaller samples [7]. In MRI classification with 2-D convolutional networks, splitting by slice rather than by subject raised a VGG16 model's accuracy on one dataset from 66% to about 97% [8]. In resting-state EEG from patients with post-traumatic stress disorder, splitting cropped trials rather than participants inflated balanced accuracy by about 5–15 percentage points for one convolutional

network and 15–20 for EEGNet [9]. In EEG, deep networks trained with segment-based rather than subject-based holdout strongly overestimated accuracy on unseen subjects, and most translational studies surveyed used segment-based holdout [25]; in EEG-based schizophrenia detection, about 65% of published studies split by epoch or selected features before partitioning, and leakage inflated accuracy by up to about 30% [26]. Neither examined how connectivity features are built. The connectome work is the closest analogue to ours, because it too derives features from pairwise connectivity; the two forms it found most damaging correspond to L5 and L1 here. Rating leakage risk from reported methods also has precedents: a survey of convolutional networks for Alzheimer's disease labeled each study's leakage status from its text [27], and a 2025 scoping review stratified 44 Alzheimer's deep-learning studies into three leakage-risk tiers and found reported accuracy highest where risk was highest [28]. We adapt that approach to the steps that build a connectivity matrix.

III. Leakage in Connectivity Pipelines

A. Pipeline formalization and connectivity measures

Let $x_i(t)$ denote the signal recorded at channel i, $i = 1, \ldots, N$, and let T be the number of samples in an epoch. A connectivity-based classifier does not operate on $x_i(t)$ directly; it operates on a matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ derived from $x_i(t)$ through a chain of steps—filtering, epoching, connectivity estimation, thresholding, normalization—each of which is fit on some subset of the data. The question this audit asks of each study is which subset that was.

Magnitude-squared coherence between channels i and j at frequency f is

$$C_{ij}(f) = \frac{|S_{ij}(f)|^2}{S_{ii}(f)\, S_{jj}(f)}, \tag{1}$$

where $S_{ij}(f)$ is the cross-spectral density [29]. S_{ij} is estimated over a window of data; if that window spans samples from both partitions, or is averaged across subjects before the partition is applied, the estimate and every feature derived from it carry information from outside the training fold.

The *phase-locking value* [30], with $\Delta\phi_{ij}(t)$ the instantaneous phase difference between channels i and j, is

$$\mathrm{PLV}_{ij} = \frac{1}{T}\left|\sum_{t=1}^{T} e^{\mathrm{j}\,\Delta\phi_{ij}(t)}\right|. \tag{2}$$

PLV is bounded in $[0, 1]$ and is usually estimated within a single epoch, which limits but does not remove the estimation route: pooling epochs across subjects before computing it still leaks.

The *phase lag index*, $\mathrm{PLI}_{ij} = \left|\left\langle \mathrm{sign}\left(\Delta\phi_{ij}(t)\right)\right\rangle_t\right|$, discounts zero-lag coupling, which volume conduction would otherwise dominate [31].

Any of these yields a weighted adjacency matrix $\mathbf{A}$. Matrices are often binarized at a threshold τ ($A^{\mathrm{bin}}_{ij} = 1$ if $A_{ij} > \tau$, else 0), or reduced to their k strongest edges. If τ or k is chosen from the distribution of A_{ij} pooled across the full dataset—a common way to fix a target graph density—it is a parameter fit on data that include the test set.

B. A taxonomy of leakage surfaces

Fig. 1 shows two orderings of the same pipeline. In the leakage-free ordering (a), the train/test partition is applied immediately after filtering, and every later step is computed separately within each fold. In the leaking ordering (b), the partition is applied last, after epoching, connectivity estimation, and threshold or normalization statistics have been computed on the pooled dataset, so the test set has already shaped the features before the model sees a split. We identify five points at which this can occur.

L1 — Epoch-level partitioning. A recording contributes many epochs, each treated as a sample. If the train/test or fold assignment is made per epoch rather than per subject, epochs from the same subject appear on both sides. *Leakage-free condition:* every epoch from a subject is assigned to the same side.

L2 — Connectivity estimation scope. Coherence, PLV, PLI, or another measure is estimated from a window of data. *Leakage-free condition:* each estimate lies entirely within one fold, and no estimate is pooled or averaged across subjects before the partition.

L3 — Threshold selection. A threshold τ or edge count k is applied to the matrix. *Leakage-free condition:* τ or k is fixed a priori or chosen from training-fold statistics only, never from a distribution pooled across the full dataset.

L4 — Feature normalization. Matrices or derived features are standardized or rescaled. *Leakage-free condition:* normalization statistics are computed on the training fold and applied unchanged to the test fold.

L5 — Channel or band selection. A subset of electrodes or frequency bands is chosen for the final model. *Leakage-free condition:* the subset is chosen a priori or from training-fold performance only, never by comparing candidate subsets on test-set accuracy.

Section IV turns these conditions into rating rules, and Table II (Section VI) into a checklist a reader can apply.

C. Why EEG subject signatures make L1 severe

L1 deserves separate emphasis because EEG carries stable subject-specific signatures: resting-state recordings, including channel coherence, identify individuals across sessions at least two weeks apart [32]. A classifier trained on epochs from a subject who also appears in the test set does not need to learn anything about depression to score well on that subject's test epochs; it can recognize the individual and recall their label. A 2026 preprint reproduced this inflation from shared subjects across five resting-state cohorts, one of them a depression cohort [33]. The literature under audit typically pairs tens of subjects (median 61, Table I) with many epochs per subject, which is the regime in which within-subject information dominates a shuffled split. The effect is compounded, not replaced, by L2–L5: even under subject-level partitioning, a connectivity or threshold estimate computed on the whole dataset can still transfer test information into the training features.

IV. Methods

A. Search and screening

The starting corpus was a previously assembled set of the 100 most-cited articles returned by a Web of Science search on EEG, connectivity, deep learning and depression **[PENDING: Q11: query string, search date and total hits if the poster team supplies them; otherwise the query reconstructed from the poster's keywords]**. Records were screened in two stages, title and abstract and then full text, against six criteria, all of which had to hold: (1) EEG is the primary modality; (2) a functional or effective connectivity measure is computed—coherence, PLV, PLI, correlation, Granger causality, mutual information, or a graph derived from one; (3) a deep learning model is applied to those features; (4) the study classifies depression or predicts its severity; (5) at least one quantitative performance metric is reported; and (6) the work is peer-reviewed, in English, and published in 2015 or later. Criterion 2 required at least one edge weight to depend on the recorded signals: a graph fixed by electrode positions or an atlas did not qualify, however the paper described it.

Because fewer than 15 Web of Science studies had qualified among the reports assessed, the protocol's supplementary search was run in OpenAlex and PubMed in September 2026. Full-text retrieval then stopped once the corpus reached 15 studies, a stopping rule the protocol did not specify, so 24 Web of Science reports that passed title and abstract screening were not assessed at full text, one of them because no copy was available to us; unobtained supplementary reports are likewise reported as not retrieved. **[PENDING: Q22: assumes option (c), closing the corpus at its current size]** Fig. 2 gives the counts and exclusion reasons at each stage.

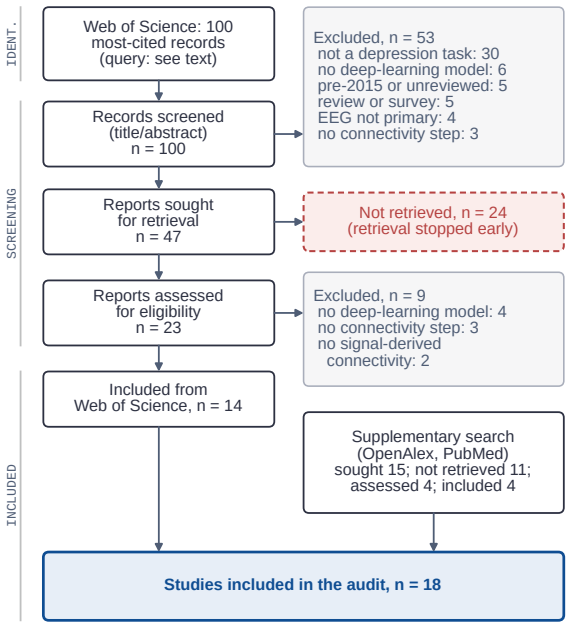


Fig. 2. Screening flow. Counts are generated from the screening log. Records in the Web of Science arm were screened on title and abstract, then on full text; the supplementary search was run when fewer than 15 studies had qualified from the Web of Science reports assessed, and retrieval stopped once the corpus reached 15 (Section IV). Web of Science assigns some records more than one document type; each of the 100 records is counted once. The dashed box marks the reports not retrieved (Section VII).

B. Extraction fields

The coding sheet's columns were fixed before any study was coded. Descriptive fields recorded the citation; the dataset (MODMA, PRED+CT, the Mumtaz resting-state dataset, a private cohort, or other); total, MDD and control sample sizes; the connectivity measure; the frequency bands; and the architecture family.

Pipeline fields recorded whether recordings were segmented into epochs and how many each subject contributed; the partitioning unit (subject, epoch, or unclear); and the validation strategy (holdout, k-fold, leave-one-subject-out, external, or unclear). Four further fields recorded, for each of surfaces L2–L5, whether the step was stated to be fit within the training fold, fit on the whole dataset, not performed, or unclear.

Performance fields recorded reported accuracy and other metrics. Where a study reported more than one accuracy, we recorded the result for the model the paper proposes under the most subject-independent protocol it reports, and its leakage fields were coded for that same protocol; where the abstract and the results disagreed, the results tables were used. This rule was fixed after the initial coding pass, which had followed each abstract's headline figure, but before any accuracy was analyzed. For three studies the raters' codes already described the most subject-independent protocol; one was left on the protocol both raters coded, with the matching accuracy, so that rating and accuracy always refer to the same protocol **[PENDING: S05: confirm with the advisor]**. The abstract's headline figure was recorded as a second field for a sensitivity analysis. Only binary depression-versus-control accuracies enter the primary comparison; other tasks enter a sensitivity analysis **[PENDING: Q20: provisional]**.

Every leakage field carries an evidence quote, the sentence on which the coding rests. A threshold, rescaling or selection step never mentioned was coded not performed; a step mentioned without saying which data it was fit on was coded unclear.

C. Rating scheme

Each study received one of four ratings, applied from reported methods alone:

- *Leakage-free*: subject-level partitioning is stated, and connectivity estimation, thresholding, normalization and channel or band selection are each stated to be confined to the training fold or not performed.
- *Leakage-possible*: subject-level partitioning is stated, but at least one of L2–L5 is not clearly described as within-fold.
- *Leakage-likely*: partitioning is at epoch level, or a preprocessing step is explicitly applied to the full dataset before partitioning.

- *Unclear*: the partitioning unit is not stated.

Unclear is a category in its own right rather than missing data, because its prevalence is itself a result for RQ1.

Raters did not assign the rating directly: they coded the partitioning unit and the four surface fields, and a fixed rule implemented in code derived the rating, so both raters' ratings follow the same logic. Where rules overlap, leakage-likely takes precedence: an explicit whole-dataset step makes a study leakage-likely whatever its partitioning unit.

Two conventions were fixed before coding and given to both raters in identical written instructions: a threshold applied to each matrix using only that matrix's own values, and normalization computed within each segment, were coded within-fold, since neither draws on other samples; and batch-normalization layers inside a network were not treated as preprocessing.

D. Rater procedure and analysis plan

Two raters coded every included study independently, without discussion, from identical written instructions. Percentage agreement and unweighted Cohen's κ on the four-level rating were computed before any disagreement was discussed. Disagreements were then resolved with a senior researcher as tiebreaker, and both pre- and post-resolution ratings are reported. A κ below 0.4 was pre-specified as indicating that the scheme is too subjective to use as defined.

The analysis was specified, and its code written, before accuracies were examined. RQ1, RQ2 and RQ4 are descriptive and no confirmatory test is performed for them. For RQ1 we report the proportion of studies in each reporting category, overall and per surface, with Wilson 95% confidence intervals. The 95% interval for κ comes from 10,000 bootstrap resamples of studies.

RQ3 is the only confirmatory analysis. We report median accuracy and interquartile range per rating stratum and compare the leakage-likely and leakage-free strata with a two-sided Mann–Whitney U test, and all strata with a Kruskal–Wallis test, each accompanied by Cliff's δ with a 10,000-resample bootstrap interval. The smallest effect of interest was fixed in advance at 3 percentage points of median accuracy; a smaller difference is reported as a null, and no subgroup was searched for an effect. Because small datasets may both favor weaker partitioning and yield higher accuracy, accuracy by rating is also reported by sample-size tercile. All resampling used a fixed seed, so every reported value can be regenerated from the coding sheet.

V. Results

A. Corpus characteristics

The audit included 18 studies: 14 from the Web of Science set and 4 from the supplementary search (Fig. 2). Table I lists each. They were published in 2020–2025. The Mumtaz resting-state dataset was used by 8 studies, MODMA by 5, PRED+CT by 2 and a private cohort by 2; 1 used another dataset. The median sample was 61 subjects (range 48–122). Convolutional or recurrent networks were used in 10 studies and graph neural networks in 6; Table I gives each study's connectivity measure. Because several studies analyzed the same public dataset and 3 came from one research group, the studies are not fully independent (Section VII).

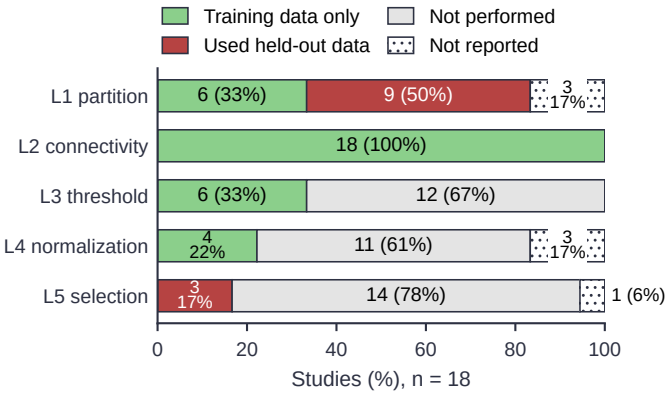


Fig. 3. Reporting completeness by leakage surface ($n = 18$ studies). Each bar is one surface; segments give the number and percentage of studies in which the step was stated to use training data only, stated to use held-out data, not performed, or not reported. For L1, these are partitioning by subject and by epoch.

B. Reporting completeness by leakage surface (RQ1)

In 22% of studies the methods did not report enough to determine whether every preprocessing step (L2–L5) was confined to the training fold (95% CI 9–45%), and 50% partitioned at the epoch level (29–71%). Fig. 3 gives the distribution for each surface.

Connectivity was estimated from each segment or subject on its own in every study (L2). Thresholds (L3) and normalization (L4) were either not used or fit per matrix or on training data, except that normalization was not described in 17%. Channel or band selection (L5) drew on all the data in 17% and was unclear in 6%. The steps specific to connectivity (L2–L4) were never reported as fit on the whole dataset; leakage risk arose instead at partitioning and at selection, and the partitioning unit and the normalization step were the ones most often left undescribed. In the secondary analysis (RQ4), studies published more recently described their pipelines less completely (Spearman ρ =-0.52, 95% CI -0.82 to -0.02), a weak trend from 18 studies.

C. Rating distribution and agreement (RQ2)

Before any discussion, the two raters assigned the same rating to 94% of studies, unweighted Cohen's $\kappa = 0.91$ (95% CI 0.66 to 1.00), above the pre-specified threshold of 0.4. On the individual fields they agreed on 17 of 18 studies for the partitioning unit, 18 for connectivity scope, 17 for thresholds, 15 for normalization and 12 for channel or band selection. Disagreement concentrated on selection, where papers compare bands or channels without saying whether the comparison chose the final model; because the rating is derived by fixed rules, most field-level disagreements left the rating unchanged.

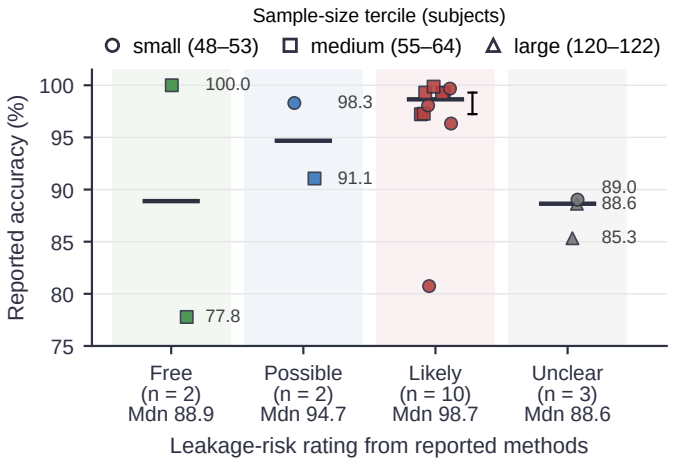


Fig. 4. Reported accuracy by leakage-risk rating for the 17 studies in the primary comparison. Each point is one study (horizontally jittered), colored by rating; horizontal bars and the values under each rating mark stratum medians; the error bar shows the interquartile range where a stratum has at least four studies; individual values are printed for strata of three or fewer. Marker shape encodes sample-size tercile.

Studies reporting several protocols were handled as described in Section IV **[PENDING: S05, deviation row 17: confirm once the advisor rules]**.

D. Accuracy by rating (RQ3)

The primary comparison used the 17 studies whose recorded accuracy came from a binary depression-versus-control task; one three-class study was excluded from it and enters only the sensitivity analysis. Median reported accuracy was 98.7% (IQR 97.2–99.3) among the 10 studies rated leakage-likely and 88.9% (IQR 83.3–94.4) among the 2 rated leakage-free, a difference of 9.8 percentage points (bootstrap 95% CI -2.8 to 21.5; Mann–Whitney $U = 10.0$, p = 1.00; Cliff's $\delta = 0.00$, 95% CI -1.00 to 1.00). Across all 4 strata, a Kruskal–Wallis test gave $H = 3.96$, p = 0.27. The difference in medians exceeds the pre-specified smallest effect of interest of 3 points, but with only 2 leakage-free studies, whose median is the midpoint of their two accuracies, it is not distinguishable from zero: the rank-based effect size is null and both intervals include no difference. This comparison is therefore uninformative. It is not evidence that leakage risk is associated with accuracy, nor evidence that it is not.

Across sample-size terciles (small, medium, large), leakage-likely studies numbered 4/6/0 and leakage-free studies 0/2/1, too few per cell to compare accuracy within terciles (Fig. 4).

The pre-specified sensitivity analyses did not change this reading. With one study per research group ($n = 16$) and without the two weakest inclusions ($n = 16$), Cliff's δ was 0.00 and 0.00; and including every task ($n = 18$) gave $\delta = 0.33$; in all three the interval for δ spanned -1 to 1. With each abstract's headline figure, only 1 study was rated leakage-free, too few for any comparison. **[PENDING: Q21: keep the analyses the advisor approves]**

VI. A Reporting Checklist

Table II lists eight items that a connectivity-based study should state. Each corresponds to one leakage surface from Section III, and each can be answered from a methods section in a sentence. Together they let a reader decide, from the paper alone, whether test information could have reached training, a question that 22% of the audited studies left open for at least one step (Section V).

Items 1 and 2 address L1; the number of epochs per subject shows how much a subject-identity shortcut could contribute under epoch-level partitioning. Items 3 and 4 address L2. Items 5 to 7 address L3 to L5, where what matters is which data a step was fit on. Item 8 bounds how far accuracy generalizes, though it does not rule out leakage within a study.

The checklist adds no new procedure: a study that already partitions by subject and fits every statistic within the training fold satisfies it with a few sentences of reporting.

VII. Discussion and Limitations

The clearest finding concerns partitioning, not the connectivity pipeline itself. Of the audited studies, 50% split recordings by epoch, which places segments from one person on both sides of the split, and a further 17% did not state the unit. None reported fitting connectivity estimation, thresholds or normalization on the whole dataset; beyond partitioning, the risk we could detect came from choosing channels or bands on all the data. What limited verification was omission: 22% of studies left at least one preprocessing step undescribed, and more recent studies described their pipelines less completely (RQ4, Section V). The accuracy comparison could not answer RQ3, because only 2 leakage-free studies reported an eligible accuracy. That absence of evidence should not be read as evidence that leakage leaves accuracy unchanged, which the neuroimaging studies in Section II show it need not, nor should any individual study's accuracy be read as inflated because of its rating.

A connectivity label is not a reliable guide to what a study computes: 2 reports assessed at full text described functional-connectivity networks but built every graph from electrode positions alone, so no edge depended on the recorded signals (excluded under criterion 2).

This audit rates reporting, not conduct. A study rated unclear or leakage-possible may have run a leakage-free pipeline and simply not said so; a study rated leakage-free may contain an error its text does not reveal. Our ratings describe what a reader can verify from the paper, which is the question that matters for how its accuracy should be read, but they are not a finding about any author's practice.

The corpus is not the product of a full systematic review. It began from a previously assembled set of the 100 most-cited Web of Science articles on the topic, whose exact query and search date could not be recovered **[PENDING: Q11: replace if the poster team supplies them]**. Citation-ranked sampling favors older and more visible work. Full-text retrieval stopped once the corpus, including studies from a supplementary search, reached the protocol's minimum of

TABLE I
CORPUS CHARACTERISTICS. ONE ROW PER INCLUDED STUDY; THE STUDY COLUMN GIVES THE REFERENCE NUMBER. THE RATING DESCRIBES WHAT THE PAPER REPORTS, NOT HOW THE STUDY WAS CONDUCTED.

Study	Dataset	N subj.	Connectivity	Architecture	Partition unit	Acc. (%)	Risk rating from reported methods
[16]	Mumtaz	64	Granger	CNN-LSTM	epoch	99.2	likely
[11]	private	48	coherence	CNN	subject	80.7	likely
[22]	Mumtaz	57	other	CNN	subject	91.1	possible
[12]	Mumtaz	64	PLV	CNN	epoch	97.3	likely
[13]	MODMA	53	PLI	CNN-LSTM	epoch	96.3	likely
[21]	PRED+CT	63	other	GCN	subject	77.8	free
[19]	Mumtaz	55	other	CNN	epoch	99.9	likely
[20]	MODMA	53	other	GCN	epoch	99.7	likely
[34]	MODMA	53	correlation	other	unclear	89.0	unclear
[35]	MODMA	53	correlation	GCN	epoch	98.1	likely
[36]	other	122	correlation	LSTM	unclear	88.6	unclear
[37]	private	67	PLI	CNN	subject	79.6[a]	free
[23]	PRED+CT	120	other	other	unclear	85.3	unclear
[14]	Mumtaz	64	other	CNN	epoch	97.2	likely
[15]	MODMA	53	correlation	GCN	subject	98.3	possible
[38]	Mumtaz	64	other	GCN	epoch	99.3	likely
[17]	Mumtaz	60	Granger	CNN	subject	100.0	free
[18]	Mumtaz	64	PLV	GCN	epoch	99.3	likely

[a] Not a binary depression-vs-control task; excluded from the primary accuracy comparison (Section V-D).

TABLE II
REPORTING CHECKLIST FOR CONNECTIVITY-BASED DEEP LEARNING STUDIES (GEN. = GENERALIZATION)

#	A connectivity study should state	Addresses
1	Whether data were partitioned by subject or by epoch	L1
2	The number of subjects and the number of epochs per subject	L1
3	Whether connectivity was estimated separately within each training fold	L2
4	The window length and overlap used for connectivity estimation	L2
5	How any threshold or edge-count parameter was selected, and from which data	L3
6	Whether feature normalization statistics were computed on training data only	L4
7	Whether channel or band selection used test-set performance	L5
8	Whether any external or cross-dataset evaluation was performed	Gen.

15 studies, a stopping rule the protocol did not specify; as a result 24 of the 47 Web of Science reports that passed title and abstract screening were not assessed (one because no copy was available to us), and 11 of the 15 supplementary reports were not retrieved. Which studies entered the audit therefore depended partly on retrieval order and access rather than on the criteria alone. **[PENDING: Q22: this sentence assumes option (c), closing the corpus at its current size]**

Several included studies analyzed the same public dataset, and 3 came from one research group with an identical sample construction, so their ratings are not fully independent; keeping one study per group did not change the result (Section V).

Both raters worked from identical written instructions and coded independently, but one is an author of this paper, and the decision rules, though mechanical, required interpretive conventions for thresholds and normalization that we fixed in advance and report in Section IV.

Availability. The coding sheet with evidence quotes, both raters' codes, and the analysis and figure code will be released on acceptance **[AUTHOR: or give an anonymized repository link]**.

VIII. CONCLUSION

We audited 18 studies applying deep learning to EEG connectivity for depression detection, rating leakage risk from reported methods alone. In 22% of them the methods did not report enough to tell whether preprocessing was confined to the training fold, and 50% partitioned by epoch. No study reported fitting connectivity estimation, thresholds or normalization on the whole dataset, and with only 2 leakage-free studies the accuracy comparison was uninformative.

A pipeline that cannot be reconstructed from its description cannot be checked for leakage. The checklist in Table II asks for eight statements that make that check possible.

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